

INSTITUTION'S INNOVATION COUNCIL MOE'S INNOVATION CELL

Institute Name:

Malla Reddy College of Engineering for women

Title of the Innovation/Prototype:

REDWINE QUALITY PREDICTION USING MACHINE LEARNING TECHNIQUES

Team Lead Name:

Jahnvi Avadhuta

Team Lead Email:

jahnavijanu666a@gmail.com

Team Lead Phone:

9392789103

Team Lead Gender:

Female

FY of Development:

2023-24

Developed as part of:

Academic Requirement/Study Project

Innovation Type:

Product,Service

TRL LEVEL:

2

Theme:

ICT, cyber-physical systems, Blockchain, Cognitive computing, Cloud computing, AI & ML.,

Define the problem and its relevance to today's market / society / industry need:

Nowadays people try to lead a luxurious life. They tend to use the things either for show off or for their daily basis. These days the consumption of red wine is very common to all. So this research basically deals with the quality prediction of the red wine using its various attributes.

Describe the Solution / Proposed / Developed:

Dataset is taken from the sources and the techniques such as Random Forest, Support Vector Machine and Naïve Bayes are applied. Various performance measures are calculated and the results are compared among training set and testing set and accordingly the best out of the three techniques depending on the training set results is predicted. Keywords— processes; data extraction; Naïve Bayes; SVM; Random Forest; quality.

Explain the uniqueness and distinctive features of the (product / process / service) solution:

Unlike traditional subjective assessments by wine tasters, this solution relies on chemical properties (such as acidity, alcohol content, and pH) as input features. These features are quantifiable and measurable, allowing for a more rigorous analysis. The solution employs a regression model to predict wine quality scores on a continuous scale (from 1 to 10). This model learns patterns from historical data, enabling it to generalize and make predictions for new wines.

How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any:

Our model is designed to handle large datasets efficiently. Whether a winery produces a few hundred or thousands of bottles, our solution scales seamlessly. Our solution not only predicts wine quality but also provides insights into which chemical properties influence the scores. This transparency helps winemakers make informed decisions.

Is there any IP or Patentable Component associated with the Solution?:

No

Has the Solution Received any Innovation Grant/Seefund Support?:

No

Are there any Recognitions (National/International) Obtained by the Solution?:

No

***Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?:**

No

Had the Solution Received any Pre-Incubation/Incubation Support?:

No

Video URL:

<https://drive.google.com/file/d/1RBZAgUjFoi26ERecb1YzYRoeFkQAyyQw/view?usp=drivesdk>

Innovation Photograph:

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